Assignment 12

Chapter 10

6. Prove Theorem 10.19.

7. For continuous random variables X and Y, discuss why $I(X;X)$ is not equal to $h(X)$.

8. Each of the following continuous distributions can be obtained as the distribution that maximizes the differential entropy subject to a suitable set of constraints:

   (a) the exponential distribution,
   $$f(x) = \lambda e^{-\lambda x}$$
   for $x \geq 0$, where $\lambda > 0$;

   (b) the Laplace distribution,
   $$f(x) = \frac{1}{2}\lambda e^{-\lambda|x|}$$
   for $-\infty < x < \infty$, where $\lambda > 0$;

   Identify the corresponding set of constraints for each of these distributions.

9. Let μ be the mean of a continuous random variable X defined on $\Re^+$. Obtain an upper bound on $h(X)$ in terms of μ.

12. *Hadamard's inequality* Show that for a positive semidefinite matrix K, $|K| \leq \prod_{i=1}^{n} K_{ii}$, with equality if and only if K is diagonal. Hint: Consider the differential entropy of a multivariate Gaussian distribution.

13. Let $K_{\mathbf{X}}$ and $\tilde{K}_{\mathbf{X}}$ be the covariance matrix and the correlation matrix of a random vector $\mathbf{X}$, respectively. Show that $|K_{\mathbf{X}}| \leq |\tilde{K}_{\mathbf{X}}|$. This is a generalization of $\mathrm{var}X \leq EX^2$ for a random variable X. Hint: Consider a multivariate Gaussian distribution with another multivariate Gaussian distribution with zero mean and the same correlation matrix.